

Algebra general exam. January 18th 2011, 9am-2pm

Directions.

- Please show all your work and justify any statements that you make.
 - State clearly and fully any theorem you use.
 - Vague statements and hand-waving arguments will not be appreciated.
 - You may assume the statement in an earlier part proven in order to do a later part.
- DO EACH PROBLEM ON A SEPARATE SHEET OF PAPER, AND STAPLE THEM TOGETHER IN THE CORRECT ORDER BEFORE TURNING THE EXAM IN.

1. **Problem 1**

Let G be a group of order 30 .

- (a) (8 pts) Prove that G has a normal subgroup of order 3 or a normal subgroup of order 5.
- (b) (6 pts) Prove that G has a normal subgroup of order 15 .

2. **Problem 2**

Let p be a prime and G a finite abelian group of p -power order. Denote by

- $e(G)$ the number of elements of order p in G
 - $s(G)$ the number of subgroups of order p in G
 - $t(G)$ the number of subgroups of index p in G
 - $h(G)$ the number of non-trivial homomorphisms from G to $\mathbb{Z}/p\mathbb{Z}$
- (a) (7 pts) Prove that $s(G) = e(G)/(p-1)$ and $t(G) = h(G)/(p-1)$
 - (b) (7 pts) Prove that $s(G) = t(G)$. Hint: By (a) this is equivalent to showing that $h(G) = e(G)$. Prove the latter by expressing both quantities in terms of r , the number of summands in the elementary divisor decomposition of G .

3. **Problem 3**

Prove that the following polynomials are irreducible in $\mathbb{Q}[x]$:

(a) (5pts) $f(x) = x^3 - 3x^2 + 15x - 7$

(b) (7 pts) $f(x) = x^5 - 3x^2 + 15x - 7$

Hint for (b): Use reduction mod 3.

Problem 4

Let R be a commutative ring with 1 , and let Ω be the set of all ideals I of R such that every element of I is 0 or a zero divisor.

(a) (5 pts) Prove that Ω has a maximal element (with respect to inclusion) and moreover any element of Ω is contained in a maximal element.

(b) (8 pts) Let I be a maximal element of Ω . Prove that I must be prime.

Problem 5

For a field F and a positive integer $n \geq 2$ denote by $M_n(F)$ the set of all $n \times n$ matrices over F .

(a) (8 pts) Assume that F is algebraically closed of characteristic NOT equal to 2 . Prove that for every invertible matrix $A \in M_n(F)$ there exists $B \in M_n(F)$ such that $B^2 = A$.

(b) (8 pts) Let F be a field of characteristic 2 . Find an invertible matrix $A \in M_n(F)$ which cannot be written as B^2 for any $B \in M_n(F)$.

Hint (for both parts): Use the Jordan Canonical Form.

Problem 6

Let R be a commutative ring with 1 . Let M, N and P be R -modules, and let $\phi : M \rightarrow N$ be an R -module homomorphism.

(a) (4 pts) Prove that there exists unique R -module homomorphism $\Phi : M \otimes_R P \rightarrow N \otimes_R P$ such that $\Phi(m \otimes p) = \phi(m) \otimes p$ for all $m \in M$ and $p \in P$.

(b) (3 pts) Assume that ϕ is surjective. Prove that Φ from part (a) is also surjective.

(c) (4 pts) Show by example that if ϕ is injective, Φ need not be injective.

Problem 7

Let p be a prime, let $\zeta_p \in \mathbb{C}$ be a primitive p^{th} root of unity and $K = \mathbb{Q}(\zeta_p)$.

(a) (7 pts) Prove that the extension $K/\mathbb{Q}$ is Galois and $\text{Gal}(K/\mathbb{Q})$ is cyclic of order $p-1$.

(b) (5 pts) Prove that K contains a unique subfield L of the form $\mathbb{Q}(\sqrt{m})$ where $m \in \mathbb{Z}$ and m is not a perfect square.

In parts (c)-(e) of this problem let $p = 7$, and let L denote the subfield found in part (b).

(c) (2 pts) Find (explicitly) a generator for the group $\text{Gal}(K/L)$.

(d) (3 pts) Find an element $\alpha \in L \setminus \mathbb{Q}$. You may express α as a polynomial in ζ_7 . Hint: use (c).

(e) (3 pts) Find m from part (b) explicitly (it is well defined up to multiplication by a perfect square).